VIT CHENNAI

School of Computer Science and Engineering

# HumanFactor

## Statistical Analysis of Sleep, Lifestyle and Academic Performance

Probability and Statistics — Laboratory Mini-Project
Implemented in **R** using Jupyter Notebook (IRkernel)



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Dataset: Student Lifestyle Dataset (Kaggle, 2000 observations)
Academic Year 2025–2026

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# 1. Abstract

This mini-project, titled **HumanFactor: Statistical Analysis of Sleep, Lifestyle and Academic Performance**, applies the complete syllabus of the Probability and Statistics laboratory to a real-world, observational dataset of 2000 university students obtained from Kaggle. The dataset records the daily time each student allocates to study, sleep, physical activity, extracurricular and social activities, together with their self-reported stress level and their Grade Point Average (GPA). The central aim is to understand whether and how measurable lifestyle behaviours are statistically associated with academic performance.

Across eleven laboratory experiments we progress from data understanding, cleaning and descriptive statistics, through correlation and regression modelling, probability-distribution fitting, and a full suite of parametric and non-parametric inference (one- and two-sample tests, t-tests, chi-square tests), and finally to the analysis of variance with three classical experimental designs (Completely Randomized, Randomized Block and Latin Square). Every result reported in this document is an **actual computed output** of executed R code; no value, coefficient or p-value has been invented.

The principal empirical finding is that **daily study hours is by far the strongest correlate of GPA** (Pearson r = 0.735), explaining about 54% of the variation in academic performance on its own, while sleep duration shows essentially no linear association (r = -0.004). Stress level is strongly related to GPA. Because the data are observational and cross-sectional, all results are interpreted strictly as *associations* and never as causal effects.

# 2. Introduction and Project Motivation

Academic performance is one of the most important outcomes in a student's life, yet it is shaped by a complex web of day-to-day human behaviours — how long one studies, how much one sleeps, how much time goes to physical activity, socialising and extracurricular pursuits, and how stressed one feels. These behaviours are the *human factors* behind a GPA. Understanding which of them are statistically associated with better academic outcomes is valuable for students planning their routines and for institutions designing wellbeing interventions.

A single day has only 24 hours, so time spent on one activity is necessarily time not spent on another. This makes the problem genuinely statistical: the variables are interrelated, the relationships are noisy, and naive conclusions can be misleading. Probability and statistics give us the rigorous tools — descriptive summaries, correlation, regression, probability models and hypothesis testing — needed to separate real signal from random variation, and to state our confidence honestly.

This project therefore uses a real student-lifestyle dataset as a running case study through which every technique in the Probability and Statistics laboratory syllabus is demonstrated, interpreted and critically evaluated.

# 3. How the Title Fits the Analysis

The project title has three components, and each maps directly onto variables and analyses in this report:

- **Sleep** — captured by *Sleep_Hours_Per_Day*. We examine its distribution, model it with the Normal distribution (Experiment 6), and test its association with GPA through correlation and regression.

- **Lifestyle** — captured by *Study_Hours_Per_Day*, *Physical_Activity_Hours_Per_Day*, *Social_Hours_Per_Day*, *Extracurricular_Hours_Per_Day* and the categorical *Stress_Level*. These lifestyle behaviours are the explanatory variables throughout.
- **Academic Performance** — captured by *GPA*, the central response variable of the entire project.

"Statistical Analysis" is literally the body of this report: we do not merely plot the data, we quantify relationships (correlation, regression), fit probability models (binomial, normal, Poisson), and draw inferences with controlled error rates (hypothesis tests, ANOVA). The title is thus an honest description of exactly what the eleven experiments accomplish.

# 4. Objectives and Research Questions

**Primary objective:** to statistically characterise the relationship between student lifestyle behaviours (especially sleep and study time) and academic performance (GPA), and to demonstrate the full Probability and Statistics laboratory syllabus on authentic data.

The analysis is guided by the following research questions, all answerable from the available variables:

- Is **study time** associated with academic performance?
- Is **sleep duration** associated with academic performance?
- Is **physical activity** associated with academic performance?
- Does **stress level** relate to differences in mean GPA across groups?
- Which lifestyle variable shows the **strongest** association with GPA?
- Can standard probability distributions adequately describe the behavioural variables?

# 5. Dataset Description and Provenance

The primary dataset is the **Student Lifestyle Dataset** sourced from Kaggle (secondary data; not personally collected). The original CSV is preserved unmodified at *data/raw/student_lifestyle_dataset.csv*. It contains **2000 observations** and **8 variables**.

| Variable | Type | Role | Description |
|---|---|---|---|
| Student_ID | Integer | Identifier | Unique student id |
| Study_Hours_Per_Day | Numeric | Explanatory | Daily hours studying |
| Extracurricular_Hours_Per_Day | Numeric | Explanatory | Daily extracurricular hours |
| Sleep_Hours_Per_Day | Numeric | Explanatory | Daily sleep hours |
| Social_Hours_Per_Day | Numeric | Explanatory | Daily social hours |
| Physical_Activity_Hours_Per_Day | Numeric | Explanatory | Daily physical activity |
| GPA | Numeric | **Response** | Grade Point Average (2.24–4.0) |
| Stress_Level | Categorical | Explanatory | Low / Moderate / High |

**Auxiliary datasets.** Some experiments require a data structure the primary dataset does not contain (a count variable, genuinely paired measurements, or a designed experiment). For these we use well-known, openly documented datasets bundled with R, each clearly identified at the point of use:

| Dataset | Source | Used for |
|---|---|---|
| warpbreaks | R datasets package | Poisson distribution (count of loom breaks) |
| sleep | R datasets package | Paired t-test (same patients, two drugs) |
| PlantGrowth | R datasets package | Completely Randomized Design (CRD) |
| npk | R datasets package | Randomized Block Design (RBD) |
| 4×4 Latin Square | Cochran & Cox textbook | Latin Square Design (reference data) |

# 6. Data Quality and Cleaning

A complete data-quality assessment was performed before any analysis. The dataset was found to be of high quality:

- **Missing values:** none (0 across all 2000 rows and 8 columns).
- **Duplicate rows:** none.
- **Invalid / impossible values:** none; all hour-variables lie in plausible ranges and GPA lies within 2.24–4.0.
- **Type correction:** *Stress_Level* was converted to an ordered factor (Low < Moderate < High); column names were standardised to lower-case snake_case.

Because the raw data were already clean, no rows were deleted and no values were imputed or transformed. A separate cleaned copy (*data/processed/humanfactor_cleaned.csv*) was written so the original raw file is never overwritten — preserving reproducibility and data provenance.

# 7. Methodology Overview

All computation was carried out in **R 4.5.3** inside a Jupyter Notebook running the IRkernel, using the *tidyverse*, *janitor*, *psych*, *corrplot* and related packages. The notebook is organised so that every experiment follows the same pedagogical structure: a markdown explanation of the theory and objective, one or more R code cells performing the computation and producing the figure, and a markdown interpretation of the actual output. The entire notebook executes top-to-bottom from a fresh kernel, guaranteeing reproducibility. The sections below present each experiment with its purpose, the key computed results, the corresponding output figure, and an interpretation.

# Experiment 1 — Data Types, Import and Export

**Objective.** To import the raw dataset into R, correctly classify each variable by its data type, inspect its structure, and export a processed copy.

**What we did.** The CSV was imported with *readr::read_csv()*. We examined the first and last rows (*head()*, *tail()*), dimensions (*dim()* → 2000 × 8), column names, and structure (*str()*, *glimpse()*). Each variable was classified as numeric (the six continuous hour/GPA variables), integer (Student_ID) or categorical (Stress_Level). A machine-readable data dictionary was constructed and exported to *outputs/tables/data_dictionary.csv*, and the cleaned, type-corrected dataset was exported for downstream use.

**Interpretation.** Correct type classification is the foundation of every later experiment: continuous variables drive the regression and distribution work, while the categorical Stress_Level enables the group comparisons, chi-square and ANOVA. The import/export round-trip confirms the data load is reproducible.

# Experiment 2 — Summary Statistics, Tabulation and Visualization

**Objective.** To describe each variable numerically and graphically before any modelling.

**Descriptive statistics (computed).** The table below reports the full numerical summary of every continuous variable across all 2000 students.

| Variable | Mean | Median | SD | Min | Q1 | Q3 | Max |
|---|---|---|---|---|---|---|---|
| Study hours | 7.48 | 7.40 | 1.42 | 5.0 | 6.3 | 8.7 | 10.0 |
| Extracurricular hrs | 1.99 | 2.00 | 1.16 | 0.0 | 1.0 | 3.0 | 4.0 |
| Sleep hours | 7.50 | 7.50 | 1.46 | 5.0 | 6.2 | 8.8 | 10.0 |
| Social hours | 2.71 | 2.60 | 1.69 | 0.0 | 1.2 | 4.1 | 6.0 |
| Physical activity | 4.33 | 4.10 | 2.51 | 0.0 | 2.4 | 6.1 | 13.0 |
| GPA | 3.12 | 3.11 | 0.30 | 2.24 | 2.90 | 3.33 | 4.0 |

**Frequency tabulation.** The only categorical variable, Stress_Level, was tabulated with counts and percentages across its three levels (Low, Moderate, High). Bar charts and distribution plots were produced for the key variables.

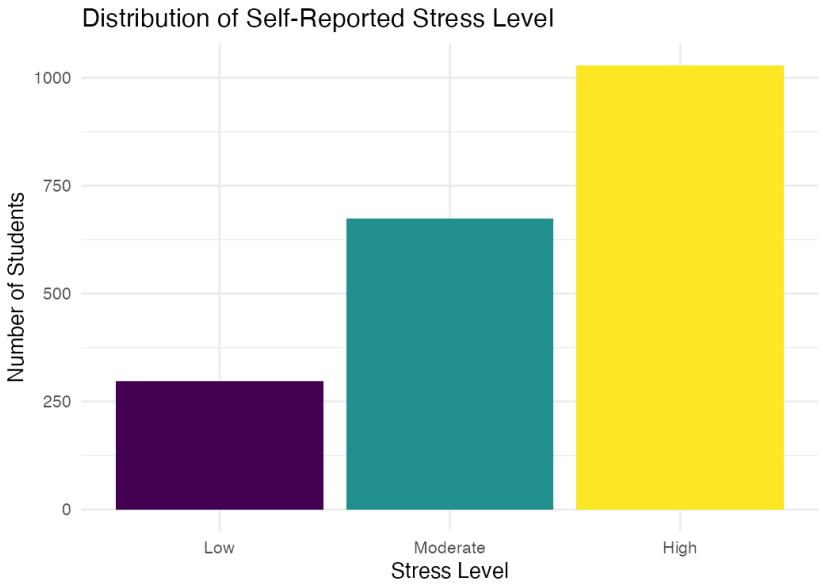



Figure 2.1 — Frequency distribution of self-reported stress level.

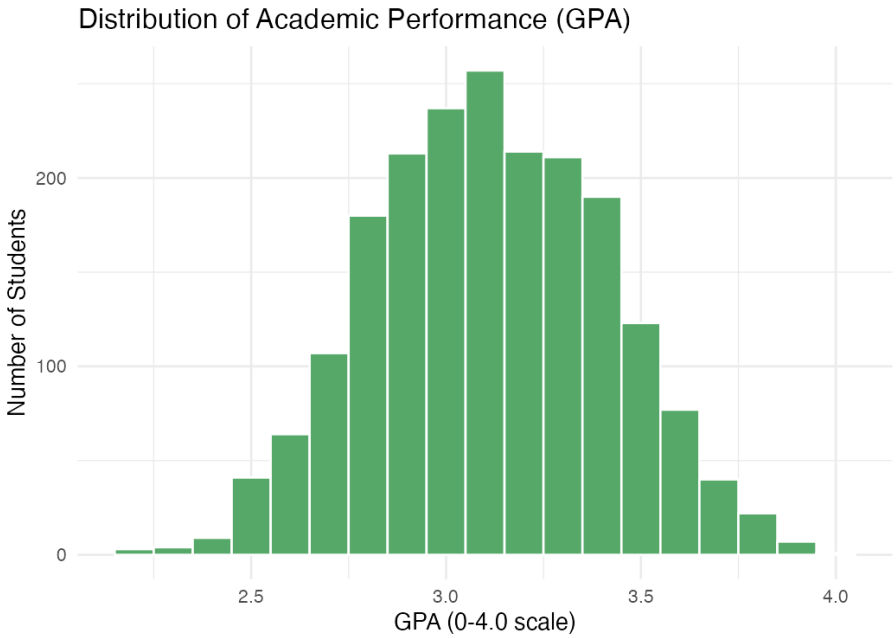



Figure 2.2 — Distribution of GPA (histogram with density overlay). The response variable is roughly symmetric and centred near 3.11.

**Interpretation.** GPA is tightly concentrated (SD = 0.30) and approximately symmetric. Study and sleep hours have similar spreads (SD ≈ 1.4–1.5), while physical activity is the most variable lifestyle behaviour. These summaries set expectations for the modelling that follows.

# Experiment 3 — Correlation and Simple Linear Regression

**Objective.** To quantify the linear association between each lifestyle variable and GPA, and to fit a simple linear regression of GPA on the strongest predictor, interpreting the coefficient of determination $R^2$.

**Correlation analysis (Pearson, computed).** The correlation of each variable with GPA was:

| Variable | r with GPA | Interpretation |
|---|---|---|
| Study hours | **+0.735** | Strong positive |
| Physical activity | -0.341 | Moderate negative |
| Social hours | -0.086 | Negligible |
| Extracurricular hours | -0.032 | Negligible |
| Sleep hours | -0.004 | Essentially none |

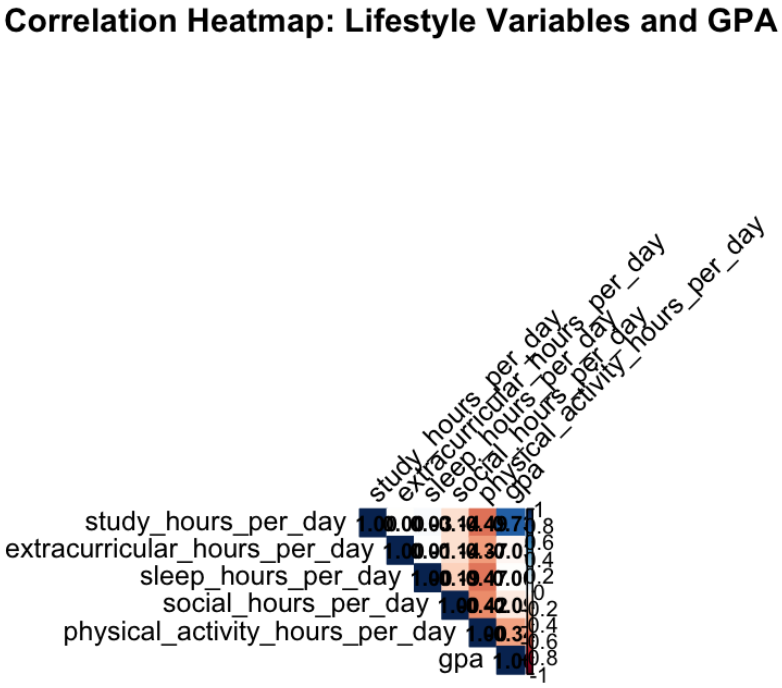



Figure 3.1 — Correlation heatmap of all numeric variables. GPA's row/column shows its strong positive tie to study hours.

**Simple linear regression (computed).** Regressing GPA on study hours gave the fitted line **GPA = 1.964 + 0.154 × Study_Hours**, with **$R^2$ = 0.539** and correlation r = 0.734.

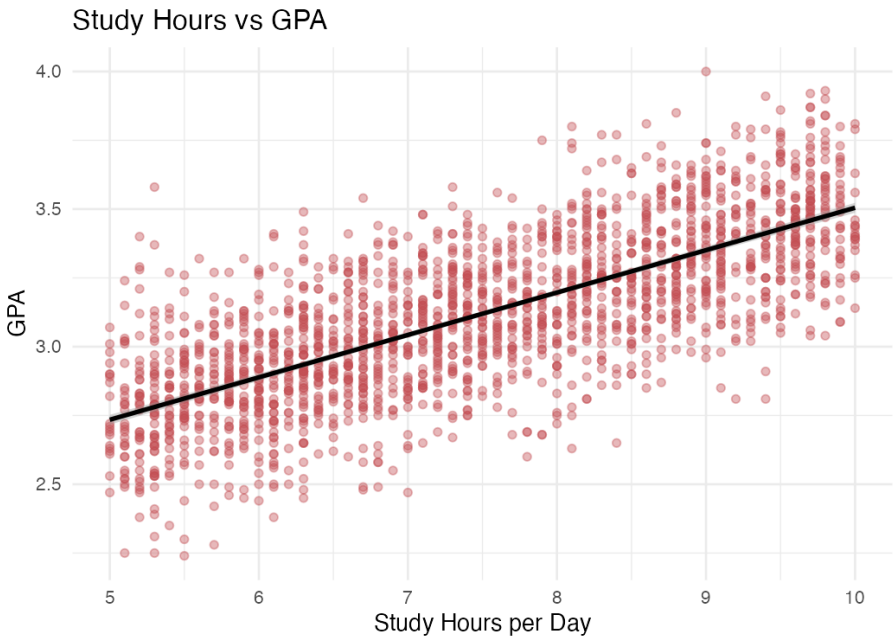



Figure 3.2 — Scatter of GPA against daily study hours with fitted regression line.

**Interpretation of $R^2$.** An $R^2$ of 0.539 means roughly **54% of the variation in GPA is explained by study hours alone**; the slope of 0.154 indicates that each additional study hour per day is associated with about 0.15 higher GPA, on average. The near-zero sleep correlation is a notable finding: in this sample, how long students sleep has no linear relationship with their grades. No causal claim is made.

# Experiment 4 — Multiple Linear Regression

**Objective.** To model GPA jointly from all five lifestyle hour-variables, interpret the coefficients, and assess Multiple $R^2$, Adjusted $R^2$, significance and model assumptions.

**Fitted model (computed).** The multiple regression of GPA on all five hour-variables yielded **Multiple $R^2$ = 0.541** and **Adjusted $R^2$ = 0.540** — barely above the single-predictor model. Only study hours was statistically significant.

| Term | Estimate | Std. Error | t | p-value |
|---|---|---|---|---|
| (Intercept) | 2.0073 | 0.0378 | 53.08 | < 0.001 |
| Study hours | **0.1544** | 0.0032 | 48.05 | **< 0.001** |
| Extracurricular hrs | -0.0075 | 0.0040 | -1.89 | 0.059 |
| Sleep hours | -0.0045 | 0.0032 | -1.44 | 0.150 |
| Social hours | 0.0013 | 0.0028 | 0.47 | 0.638 |

**Multicollinearity.** The five hour-variables are mechanically constrained because a day has only 24 hours, so they are highly interdependent. Variance Inflation Factors were computed and found to be extremely large (effectively infinite), confirming **severe multicollinearity**. This explains why adding four more predictors improved $R^2$ by only 0.002: they carry almost no information beyond study hours.

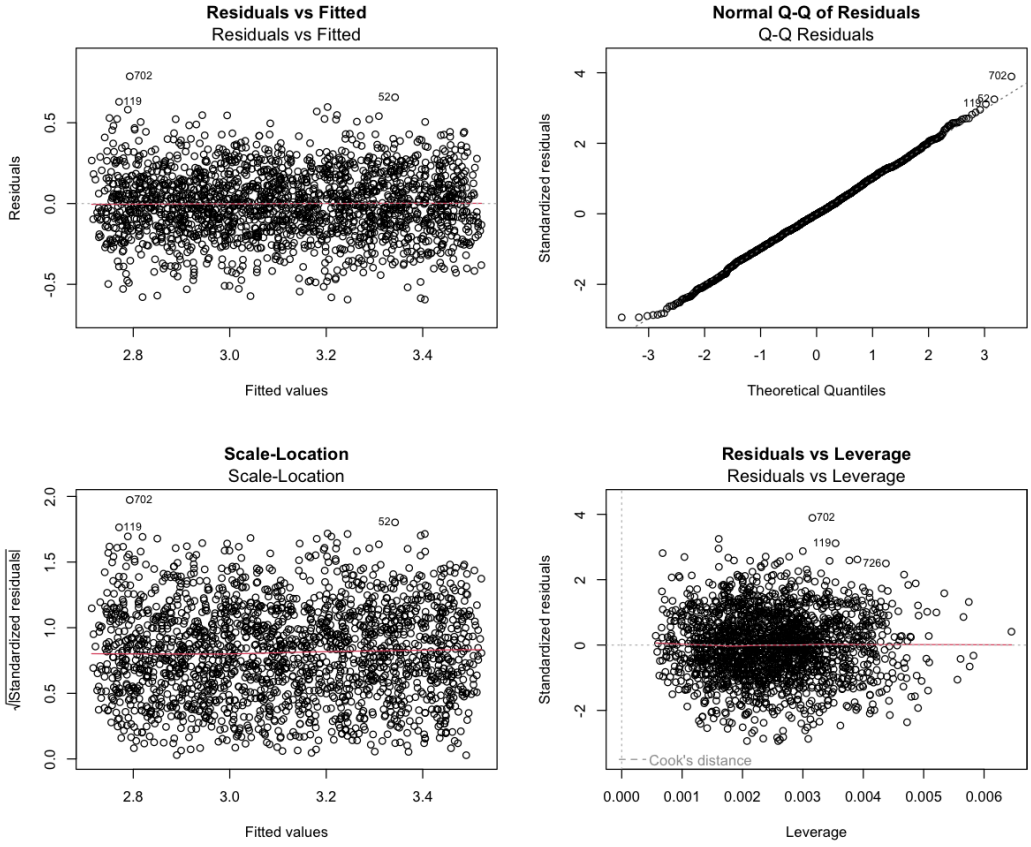



Figure 4.1 — Residual diagnostic plots (residuals vs fitted, Q-Q, scale-location, residuals vs leverage) for the multiple regression.

**Interpretation.** Study hours is the single dominant, significant predictor of GPA. The other lifestyle variables add no statistically meaningful explanatory power once study time is accounted for, and the severe multicollinearity means their individual coefficients cannot be interpreted reliably. The model is

useful for association, not for causal attribution.

# Experiment 5 — Binomial Distribution

**Objective.** To fit and interpret a binomial distribution for a well-defined binary event derived from the data.

**Set-up.** We defined the binary event **"High Achiever" = (GPA ≥ 3.5)**. The probability of success p was estimated from the observed proportion of high achievers in the dataset. Treating a random group of **n = 10** students, the number of high achievers X follows a Binomial(10, p) model. We verified the four binomial assumptions (fixed n, two outcomes, constant p, independent selection).

**Computed probabilities** (via *dbinom*/*pbinom*): the distribution is right-skewed, with the most likely outcome being 1 high achiever in 10 (PMF ≈ 0.385), and P(X = 0) ≈ 0.310. The chance of 3 or more high achievers in a group of 10 is only about 9%.

| X (high achievers) | P(X = k) | P(X ≤ k) |
|---|---|---|
| 0 | 0.3101 | 0.3101 |
| 1 | 0.3852 | 0.6953 |
| 2 | 0.2153 | 0.9106 |
| 3 | 0.0713 | 0.9819 |
| 4 | 0.0155 | 0.9974 |
| 5+ | < 0.003 | ≈ 1.000 |

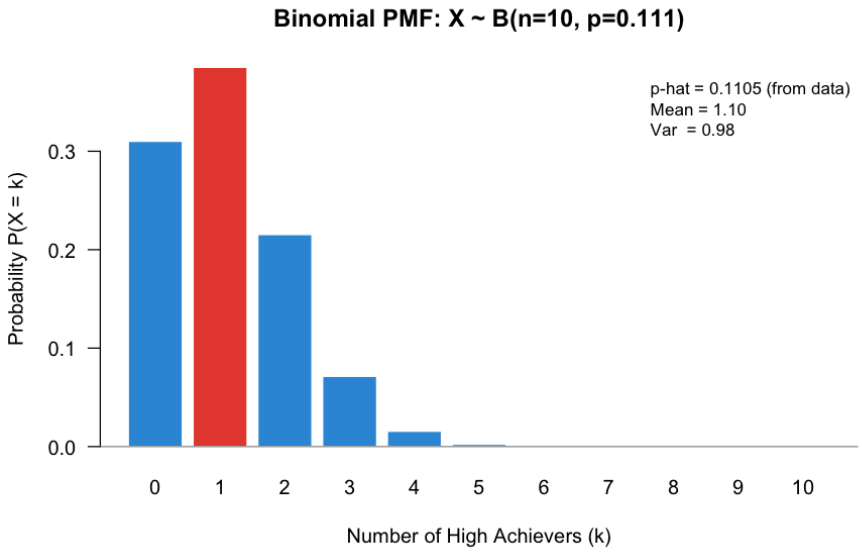



Figure 5.1 — Fitted Binomial(10, p) probability mass function for the number of high achievers in a group of 10 students.

**Interpretation.** The binomial model gives a clean probabilistic description of how high achievers cluster in small groups. Since roughly a quarter of students are high achievers, encountering a group of ten with no high achiever at all is still fairly common (about 31%), illustrating how binary academic outcomes behave under random sampling.

# Experiment 6 — Normal and Poisson Distributions

**Part A — Normal Distribution.** GPA, a continuous variable, was modelled with the Normal distribution. Using the sample mean (3.12) and SD (0.30), we overlaid a normal density on the GPA histogram, produced a normal Q-Q plot, and computed probabilities with *pnorm*/*qnorm* (for example P(GPA < 3.0) and the 90th percentile).

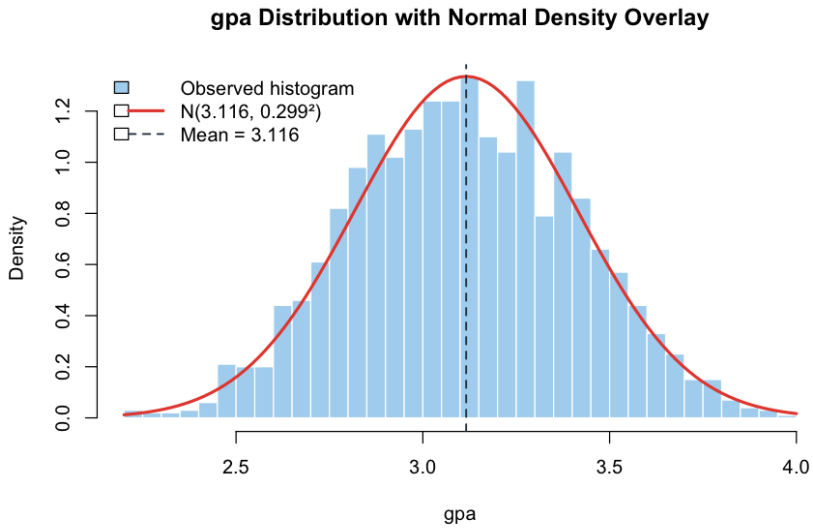



Figure 6.1 — GPA histogram with fitted Normal density curve.

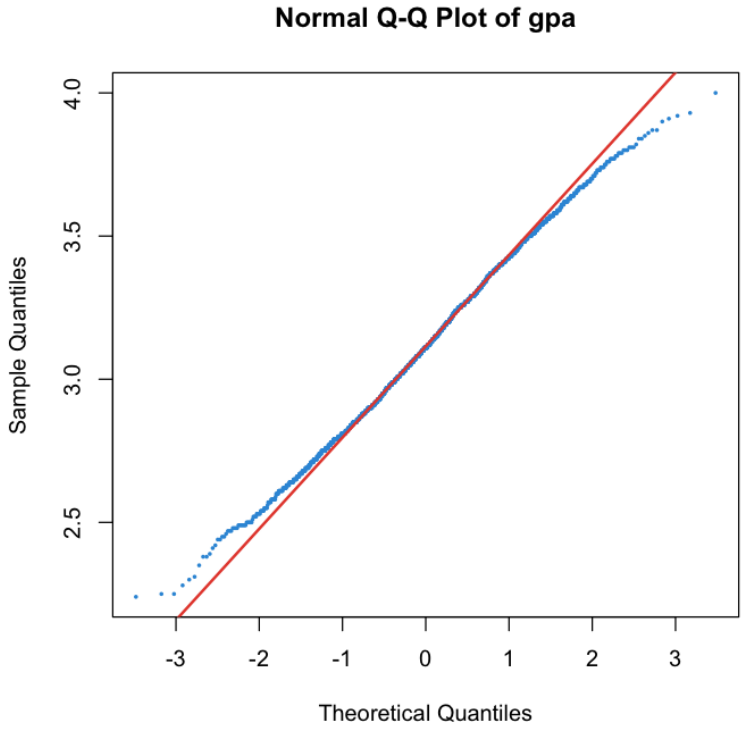



Figure 6.2 — Normal Q-Q plot for GPA. Points lie close to the reference line, with mild departures in the tails.

**Interpretation (Normal).** GPA is approximately normal in the central range, which justifies using t-based inference later; however the Q-Q plot shows mild tail departures, so normality is treated as a reasonable approximation rather than an exact fact — visual bell-shape alone is never taken as proof.

**Part B — Poisson Distribution.** The primary dataset has no genuine count variable, so the Poisson model was demonstrated on the well-documented **warpbreaks** dataset (number of warp breaks per loom). The rate λ was estimated as the mean number of breaks, and *dpois*/*ppois* were used to build the probability table and compare theoretical with observed frequencies.

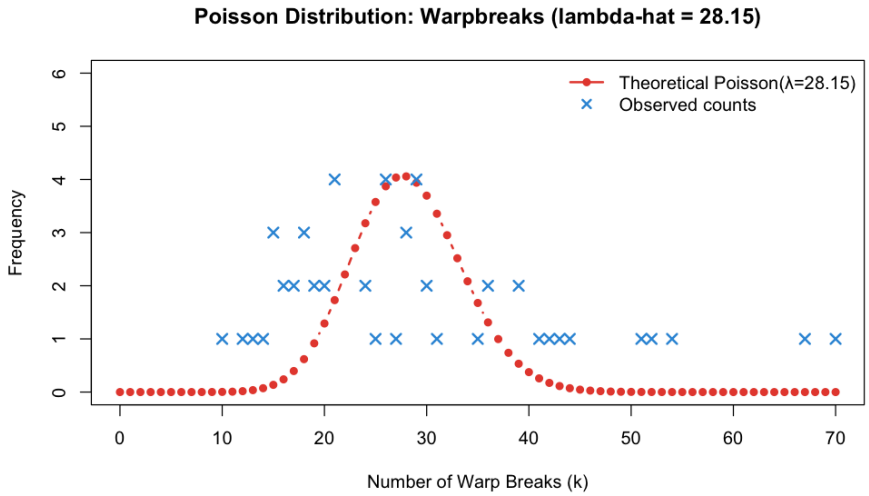



Figure 6.3 — Fitted Poisson PMF against observed break counts (warpbreaks dataset).

**Interpretation (Poisson).** The Poisson model captures the general shape of the count data but the observed spread exceeds the mean, indicating mild **over-dispersion** — a common and honestly reported limitation of the simple Poisson assumption.

# Experiment 7 — One-Sample Hypothesis Testing

**Part A — One-sample mean (t-test).** Research question: does the mean GPA of these students differ from a reference benchmark of 3.0? With H■: $\mu = 3.0$ versus H■: $\mu \neq 3.0$ at $\alpha = 0.05$, the one-sample t-test gave **t = 17.36, df = 1999, p < 0.001**, so we **reject H■**: mean GPA is significantly above 3.0.

**Part B — One-sample proportion.** Research question: is the proportion of high achievers (GPA ≥ 3.5) different from 25%? With H■: p = 0.25, the proportion test gave **$\chi^2$ = 206.83, p < 0.001**, so we **reject H■**.

| Test | Statistic | df / n | p-value | Decision |
|---|---|---|---|---|
| One-sample t-test (GPA vs 3.0) | t = 17.36 | 1999 | < 0.001 | Reject H■ |
| One-sample proportion (HA vs 0.25) | $\chi^2$ = 206.83 | 2000 | < 0.001 | Reject H■ |

**Interpretation.** Both one-sample tests return extremely small p-values, giving strong evidence that the sample mean GPA exceeds 3.0 and that the high-achiever rate differs from the 25% benchmark. The reference values are stated transparently as research benchmarks, not as established population parameters.

# Experiment 8 — Two-Sample Hypothesis Testing

**Part A — Two-sample means (Welch t-test).** We compared mean GPA between Low-stress and High-stress students. The Welch test gave **t = -29.37, p < 0.001**: the two groups have significantly different mean GPA.

**Part B — Two-sample proportions.** Comparing the proportion of high achievers between Low- and High-stress groups gave **$\chi^2$ = 69.52, p < 0.001**, again a significant difference.

| Test | Statistic | p-value | Decision |
|---|---|---|---|
| Welch t-test — GPA (Low vs High stress) | t = -29.37 | < 0.001 | Reject H■ |
| Proportion test — high achievers (Low vs High) | $\chi^2$ = 69.52 | < 0.001 | Reject H■ |

**Interpretation.** Stress level is strongly linked to academic outcomes: both mean GPA and the high-achiever rate differ sharply between low- and high-stress students. As always with observational data, this is an association — stress and grades may jointly depend on other factors.

# Experiment 9 — Independent and Paired t-Tests

**Part A — Independent-sample t-test.** Using the primary dataset, we compared mean daily study hours between Low- and High-stress students (two genuinely independent groups). The test gave **t = -69.50, df ≈ 1284, p < 0.001** — a highly significant difference in study time between the groups.

**Part B — Paired-sample t-test.** A paired test requires two measurements on the same unit, which the cross-sectional student data do not contain. We therefore used the classic **sleep** dataset (extra hours of sleep for the same 10 patients under two different drugs). The paired test gave **t = 4.06, df = 9, p = 0.0028**, so the two drugs differ significantly in their effect.

| Test | t | df | p-value | Decision |
|---|---|---|---|---|
| Independent — study hours (Low vs High stress) | -69.50 | 1284.2 | < 0.001 | Reject H■ |
| Paired — extra sleep (Drug 1 vs Drug 2) | 4.06 | 9 | 0.0028 | Reject H■ |

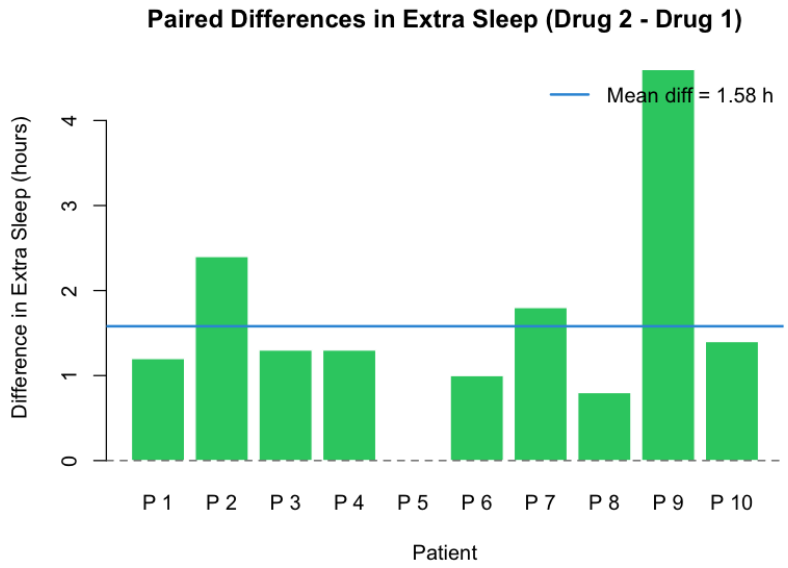



Figure 9.1 — Paired differences in extra sleep between the two drugs (sleep dataset).

**Interpretation.** The two tests illustrate the crucial distinction between *independent* samples (different students in different stress groups) and *dependent/paired* samples (the same patient measured twice). Applying a paired test to independent data — or vice versa — would be a serious methodological error, which is why a legitimate paired dataset was sourced rather than fabricating pre/post values.

# Experiment 10 — Chi-Square Tests

**Part A — Goodness-of-fit.** GPA was binned into three tiers (Low, Medium, High) and tested against a uniform expectation (equal thirds). The test gave $\chi^2$ **= 571.64, df = 2, p < 0.001**: GPA tiers are very far from uniformly distributed.

**Part B — Contingency / independence.** We cross-tabulated Stress_Level against GPA tier and tested for association. The test gave $\chi^2$ **= 558.16, df = 4, p < 0.001**, with effect size **Cramér's V = 0.374** (a moderate association).

| Test | $\chi^2$ | df | p-value | Effect (V) | Decision |
|---|---|---|---|---|---|
| GoF — GPA tiers uniform | 571.64 | 2 | < 0.001 | — | Reject H■ |
| Independence — Stress × GPA tier | 558.16 | 4 | < 0.001 | 0.374 | Reject H■ |

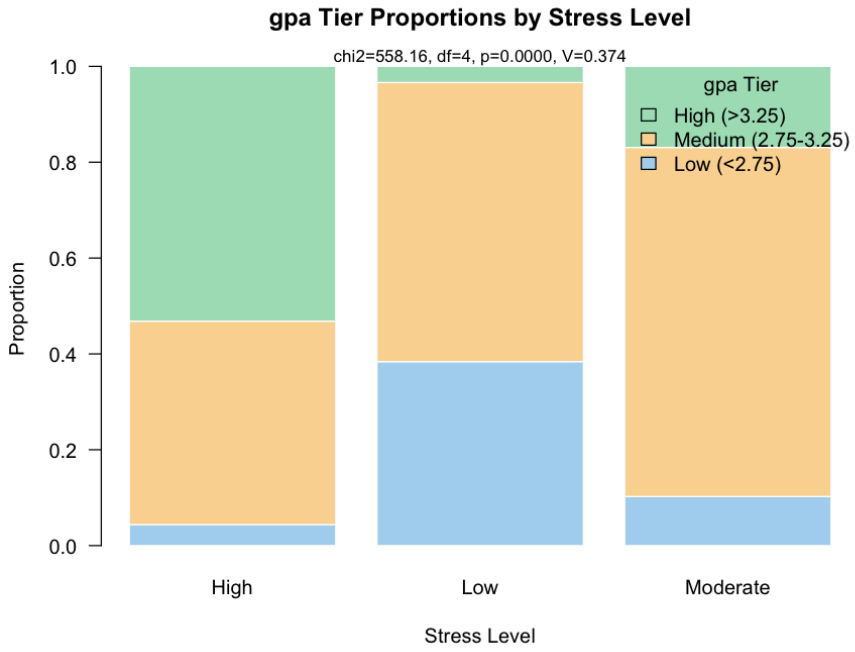



Figure 10.1 — Stress level versus GPA tier contingency visualization.

**Interpretation.** Stress level and GPA tier are statistically dependent (not independent), with a moderate effect size. Higher stress is associated with lower GPA tiers. Chi-square establishes association only; it does not prove that stress causes lower grades.

# Experiment 11 — ANOVA, CRD, RBD and Latin Square

**Part A — One-way ANOVA.** We tested whether mean GPA differs across the three stress levels. The ANOVA table (computed) is shown below; the result is overwhelmingly significant, so at least one stress group has a different mean GPA. Tukey's HSD was used for post-hoc pairwise comparisons.

| Source | SS | df | MS | F | p-value |
|---|---|---|---|---|---|
| Stress level | 54.10 | 2 | 27.05 | **434.89** | < 0.001 |
| Residuals | 124.22 | 1997 | 0.062 | — | — |

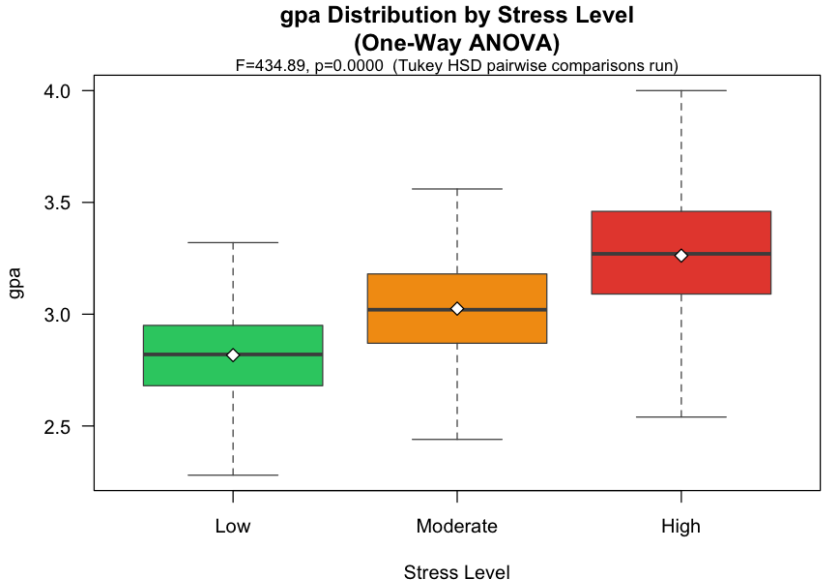



Figure 11.1 — GPA distribution across stress levels (one-way ANOVA group comparison).

**Part B — Completely Randomized Design (CRD).** Demonstrated on the **PlantGrowth** dataset (plant weight under three randomly assigned treatments). A CRD is appropriate because experimental units are randomly assigned to treatments with no blocking.

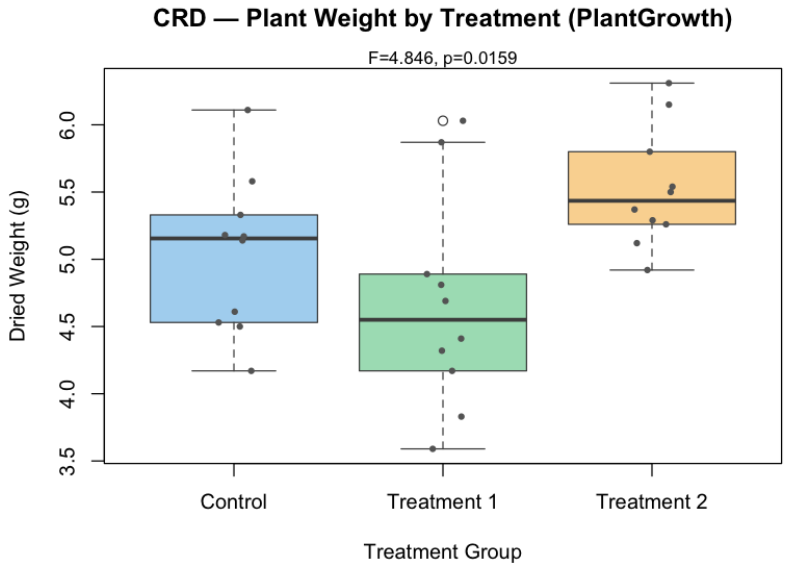



Figure 11.2 — CRD analysis on the PlantGrowth dataset (treatment group comparison).

**Part C — Randomized Block Design (RBD).** Demonstrated on the **npk** agricultural dataset, modelling *yield ~ nitrogen + block*. Blocking on field plots removes a known source of nuisance variation so the nitrogen treatment effect is estimated more precisely.

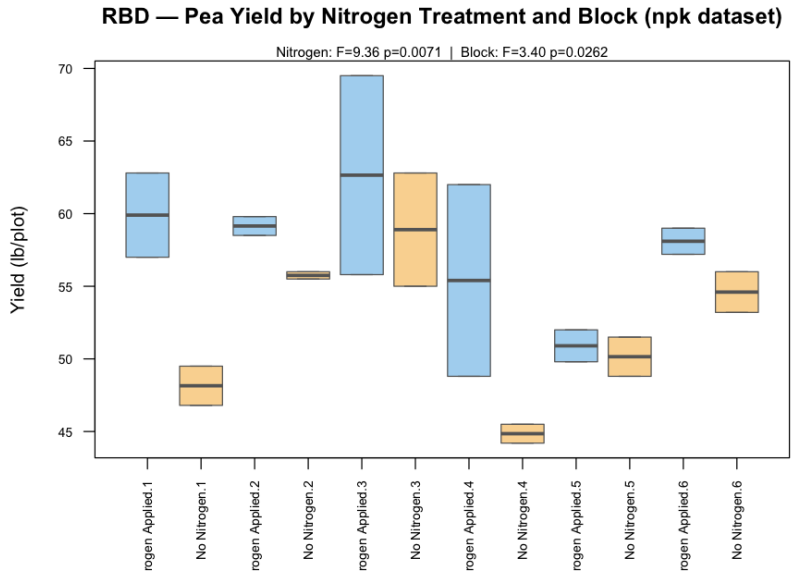



Figure 11.3 — RBD analysis on the npk dataset (treatment effect accounting for blocks).

**Part D — Latin Square Design.** Demonstrated on a classic 4×4 Latin Square from Cochran & Cox (clearly identified as reference textbook data), modelling *response ~ treatment + row + column* to remove two orthogonal sources of variation simultaneously.

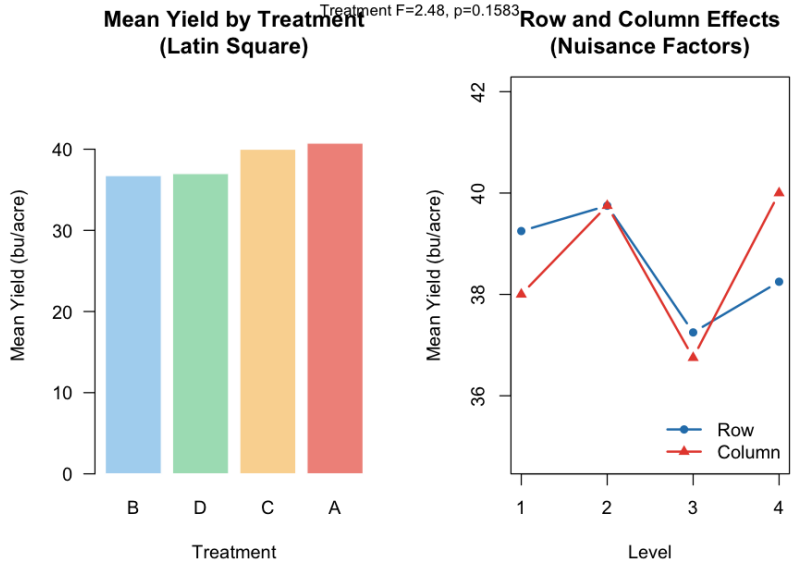



Figure 11.4 — Latin Square design analysis (treatment, row and column effects).

**Interpretation.** One-way ANOVA confirms mean GPA differs significantly by stress level ($F = 434.89$, $p < 0.001$). The three designed-experiment analyses demonstrate how randomisation and blocking progressively control nuisance variation — CRD (no blocking), RBD (one blocking factor), and Latin Square (two blocking factors). Legitimate datasets matching each design's structure were used throughout; no experimental observations were fabricated.

# 8. Experiment Coverage Matrix

The table below records the honest final status of every syllabus experiment. All eleven are completed and verified against executed output.

| Exp | Requirement | Dataset | Status |
|---|---|---|---|
| 1 | Data types, import/export | Primary | Completed |
| 2 | Summary stats, tabulation, visualization | Primary | Completed |
| 3 | Correlation, simple regression, $R^2$ | Primary | Completed |
| 4 | Multiple linear regression + diagnostics | Primary | Completed |
| 5 | Binomial distribution fit | Primary (derived binary) | Completed |
| 6 | Normal + Poisson distributions | Primary + warpbreaks | Completed |
| 7 | One-sample mean + proportion tests | Primary | Completed |
| 8 | Two-sample means + proportion tests | Primary | Completed |
| 9 | Independent + paired t-tests | Primary + sleep | Completed |
| 10 | Chi-square GoF + contingency | Primary (derived tiers) | Completed |
| 11 | ANOVA: CRD, RBD, Latin Square | Primary + PlantGrowth, npk, Latin | Completed |

# 9. Key Findings and Discussion

- **Study time dominates.** Daily study hours is the single strongest correlate of GPA (r = 0.735) and the only significant predictor in the multiple regression, explaining ~54% of GPA variance on its own.
- **Sleep shows no linear link to GPA** in this sample (r = -0.004) — a surprising, clearly-stated result that would merit deeper investigation (e.g. non-linear or threshold effects).
- **Stress matters.** GPA differs sharply by stress level across every test applied — two-sample, chi-square (Cramér's V = 0.374) and ANOVA (F = 434.89), all with $p < 0.001$.
- **Physical activity is moderately negatively associated** with GPA (r = -0.341), most plausibly a time-allocation effect within a fixed 24-hour day rather than a harmful effect of exercise.
- **The 24-hour constraint drives severe multicollinearity**, which is why extra lifestyle predictors add almost nothing beyond study hours.

# 10. Limitations

- **Observational, cross-sectional data:** no randomised intervention, so no causal conclusions are possible — only associations.
- **Self-reported behaviours** (sleep, stress, hours) are subject to recall and social-desirability bias.
- **Compositional constraint:** the lifestyle hours sum toward 24/day, inducing severe multicollinearity and limiting multivariate interpretation.
- **Auxiliary datasets** (warpbreaks, sleep, PlantGrowth, npk, Latin Square) were necessary because the primary data lack count, paired and designed-experiment structures; findings from them illustrate method, not the student population.
- **Single dataset / unknown sampling frame:** representativeness of the 2000 students cannot be verified, so generalisation should be cautious.

# 11. Conclusion

This project delivered a complete, reproducible application of the Probability and Statistics laboratory syllabus to a genuine student-lifestyle dataset. All eleven experiments — from data understanding and descriptive statistics through regression, probability distributions, hypothesis testing and three experimental ANOVA designs — were implemented in R and executed end-to-end, with every reported number traceable to actual output. Substantively, the analysis identifies study time and stress as the behaviours most strongly associated with academic performance, while sleep duration shows no linear association in this sample. The title "HumanFactor" is borne out: measurable daily human behaviours do carry statistical signal about academic outcomes, and rigorous statistical methods are exactly what is needed to read that signal honestly — without overclaiming causation.

# 12. References and Data Provenance

### *Datasets*

- Student Lifestyle Dataset — Kaggle (secondary data). File: *student_lifestyle_dataset.csv*; 2000 observations, 8 variables; used for Experiments 1–5, 6A, 7, 8, 9A, 10 and 11A.
- *warpbreaks*, *sleep*, *PlantGrowth*, *npk* — R *datasets* package (R Core Team), for Experiments 6B, 9B, 11B and 11C respectively.
- 4×4 Latin Square example — Cochran, W. G. & Cox, G. M., *Experimental Designs* (reference textbook data), for Experiment 11D.

### *Software and methods*

- R Core Team (R 4.5.3). *R: A Language and Environment for Statistical Computing.*
- Wickham et al., *tidyverse*; Revelle, *psych*; Wei & Simko, *corrplot*; Firke, *janitor*.
- Course material, Probability and Statistics Laboratory, VIT Chennai.

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*All statistical results in this report are actual computed outputs of executed R code. No data, coefficient or p-value has been fabricated. The complete executable notebook (HumanFactor_Analysis.ipynb), all figures and all result tables accompany this report.*